

Homework 1

1. (60%) Three-Asset Portfolio

Let $R_{pt} = w_1 R_{1t} + w_2 R_{2t} + w_3 R_{3t}$,

where R_{pt} is the return of portfolio at time t

w_i is the weight of investment in asset (company) i

R_{it} is the return of asset (company) i at time t

The expected return of the portfolio is $E(R_{pt}) = \mu_p = w_1 \mu_1 + w_2 \mu_2 + w_3 \mu_3$ and the variance of the portfolio is

$$Var(R_{pt}) = \sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + w_3^2 \sigma_3^2 + 2w_1 w_2 \sigma_{12} + 2w_2 w_3 \sigma_{23} + 2w_1 w_3 \sigma_{13}$$

where μ_i is the mean of asset (company) i

σ_i^2 is the variance of asset (company) i

σ_{ij} is the covariance between asset i and j

(1) (25%) Please show **in detail** that the expected return of the portfolio

$$E(R_{pt}) = \mu_p = W^T \mu \text{ and the variance of the portfolio } Var(R_{pt}) = \sigma_p^2 = W^T \Sigma W$$

$$\text{where } \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix}; W = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}; \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix}$$

(2) (10%) Given the variance-covariance matrix

$$\Sigma = \begin{bmatrix} 24 & -10 & 25 \\ -10 & 50 & 32 \\ 25 & 32 & 12 \end{bmatrix}$$

Calculate the variance of an equally weighted portfolio. Please verify your answers (by Matlab or other method).

(3) (25%) **Matrix derivative with respect to W:** Please show **in detail** that

$$\frac{\partial \sigma_p^2}{\partial W} \equiv \begin{pmatrix} \frac{\partial \sigma_p^2}{\partial w_1} \\ \frac{\partial \sigma_p^2}{\partial w_2} \\ \frac{\partial \sigma_p^2}{\partial w_3} \end{pmatrix} = \frac{\partial (W^T \Sigma W)}{\partial W} = 2\Sigma W$$

2. Covariance between Two Portfolios

(40%) Suppose we want to express the covariance between two portfolios, p and q , using matrix notation. Let $W_p^T = [w_{1p} \quad w_{2p}]$ be the weights held in portfolio p . For example, one might construct portfolio p by holding 50% of his/her wealth in asset 1 and the remaining 50% in asset 2. Similarly, let $W_q^T = [w_{1q} \quad w_{2q}]$ be the weights used to construct portfolio q .

Please show **in detail** that the covariance between portfolios p and q can be expressed as the following matrix form:

$$\text{Cov}(R_{pt}, R_{qt}) \equiv \sigma_{pq} = E[(R_{pt} - E(R_p))(R_{qt} - E(R_q))] = W_p^T \Sigma W_q$$

$$\text{where } W_p = \begin{bmatrix} w_{1p} \\ w_{2p} \end{bmatrix}; W_q = \begin{bmatrix} w_{1q} \\ w_{2q} \end{bmatrix}; \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}$$